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LSAPy: Land Suitability Analysis in Python

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Summary

LSAPy is a highly customizable Python library designed to streamline and enhance Land Suitability Analysis (LSA) workflows. The package implements a fuzzy-logic approach and provides two core objects - `SuitabilityCriteria` and `LandSuitabilityAnalysis` - and one module (`lsapy.standardize`) that work together to deliver a flexible and user-defined LSA framework. By relying on `xarray` objects for computation (Hoyer & Hamman, 2017), LSAPy seamlessly integrates with the broader Python ecosystem, such as `dask` for efficient parallel processing and `matplotlib` for data visualisation. Its modular design addresses some limitations of existing LSA tools by offering greater flexibility, reproducibility, and scalability for research and practical applications.

Statement of need

In the past decades, several software programs have been developed to perform land evaluation or suitability analysis, including ALES (Johnson & Cramb, 1991), Micro-LEIS (D. De La Rosa et al., 1992; D. De La Rosa et al., 2004), LEIGIS (Kalogirou, 2002), ALSE (Elsheikh et al., 2013) and general-purpose GIS platforms such as ArcGIS and QGIS. While each of these tools offers distinct advantages, a limitation that often arises is the software-imposed constraints and the lack of freedom left to users (Asaad et al., 2022; Chen et al., 2022; Elsheikh et al., 2013). These limitations manifest in several ways. Desktop GIS solutions, for example, often present challenges related to operating system dependency, the cost of proprietary software (e.g., ArcGIS), and difficulties in integration with broader analytical frameworks (Chen et al., 2022). Furthermore, many specialised programs (e.g., Micro-LEIS, LEIGIS, ALSE) do not permit modification of the land characteristics used in the analysis (Asaad et al., 2022; Elsheikh et al., 2013). This rigidity is problematic, as the predefined land characteristics may not apply to all crops or may require augmentation with additional parameters. Additionally, some tools (e.g., LEIGIS) support only a limited selection of crops for evaluation (Elsheikh et al., 2013).

Recently, two libraries – ALUES (Asaad et al., 2022) and PyLUSAT (Chen et al., 2022) – have been developed to address some of these limitations. ALUES, an R package, supports the evaluation of 56 crops and allows the addition of new ones. While it addresses some of the limitations discussed above, it falls short in others. For example, it relies on fixed criteria grouped into three unmodifiable categories (Asaad et al., 2022). PyLUSAT, a Python package, enables land suitability analysis using vector data, which offers specific advantages (Chen et al., 2022). Nevertheless, raster data model have been recognized as more appropriate for LSA applications because of its area-oriented structure (Malczewski, 2004). Moreover, raster-based approaches are generally more efficient for data combinations and complex calculations (Carr & Zwick, 2007), particularly when integrating climate data, which is often distributed in netCDF format and best processed using raster routines. Converting such data to a vector format for analysis is technically feasible but suboptimal, leading to increased computational overhead and memory usage. Consequently, raster-based methods are essential for large-scale analyses, such as land suitability assessments using climate projections.

Beyond these technical constraints, the lack of user flexibility in existing software introduces further limitations. First, when programs offer a restricted set of functionalities, they hinder the reproducibility of scientific results. For instance, ALSE aggregates criteria suitability using the maximum limitation method (Elsheikh et al., 2013), whereas ALUES provides only minimum, maximum, and average aggregation options (Asaad et al., 2022). This discrepancy renders the reproduction of cross-software results impossible. Second, tools designed for specific use cases (e.g., agriculture) limit broader adoption and, in the case of free and open-source software (FOSS), may impede community engagement and growth. Finally, rigid frameworks restrict the ability to explore edge cases or out-of-the-box analyses, which are increasingly important in a rapidly changing environment.

The limitations of existing software programs motivated the development of LSAPy (Land Suitability Analysis in Python), a new, highly customizable Python package that supports raster-like data. Thanks to its custom-based approach, frameworks used in most software programs mentioned here can be implemented in LSAPy.

Land Suitability Analysis Workflow

LSAPy provides two core objects and one module that operate together to perform the land suitability analysis (LSA) according to user-defined frameworks (Figure 1).

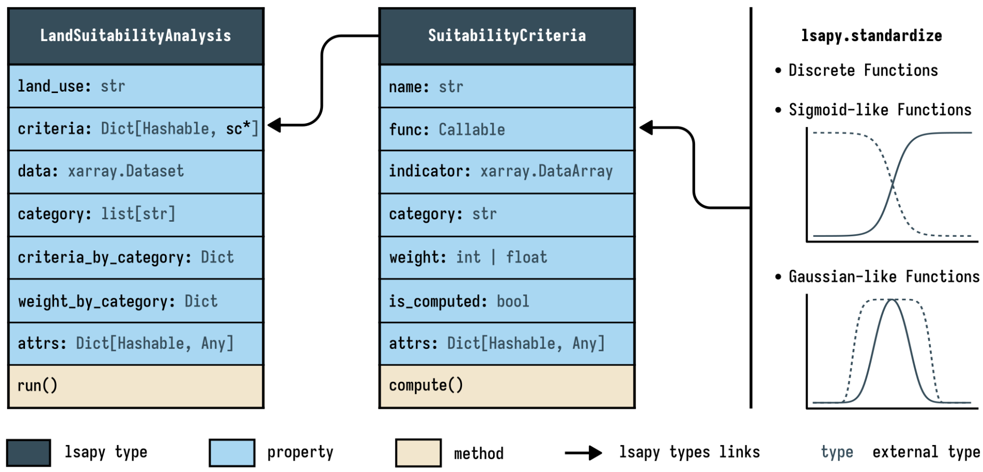

Figure 1: Overview of LSAPy's object structures and their associated properties and methods. *sc is used as an abbreviation of `SuitabilityCriteria`.

Standardize Module

`lsapy.standardize` provides a set of standardization functions used to transforms input data into standardized scale of suitability values ranging from 0 to 1 (Figure 1). LSAPy includes built-in functions for both discrete and continuous data. For the latter, the package follows a fuzzy-logic approach, implementing sigmoid-like and Gaussian-like membership functions previously used in LSA studies.

Suitability Criteria

The `SuitabilityCriteria` defines an individual criteria used in LSA. Its `func` property refers to the standardization function, while `indicator` specifies the associated input data (Figure 1).

The `weight` and `category` properties allow users to customise how each criteria is aggregated with others in the analysis. `is_computed` indicates whether the indicator already corresponds to suitability values and does not need further computation. Finally, the `compute()` method applies the standardization function to the given `indicator` to calculate the suitability score.

Land Suitability Analysis

`LandSuitabilityAnalysis` is the top-level class in LSAPy, defining the LSA framework. All criteria for the analysis are stored in the `criteria` property (Figure 1). The `category`, `criteria_by_category` and `weights_by_category` properties are populated based on the defined criteria with the first one listing all unique categories, and the two latter their associated criteria and weights respectively.The `run()` method executes the LSA, with parameters specifying the level of suitability to compute (i.e., criteria, category, or overall land suitability) and the aggregation method to use. Currently, supported aggregation methods include median, mean, weighted mean, geometric mean, weighted geometric mean, and limiting factor.

Additional Features

- The `stats` module offers functions for computing spatial (e.g., national, regional…) summary statistics of land use suitability.
- LSAPy's `open_data` function provides access to sample datasets, including soil/land data from NZGLID (Hamon, 2025) and climate data from NEX-GDDP-CMIP6 (Thrasher et al., 2022) for tutorial and training purposes.

Research Applications

LSAPy has been used to assess the impact of climate change on apples, cherries, maize and wheat in New Zealand (Hamon, Quénol, et al., 2025a; Hamon, Quénol, et al., 2025b), and is the core component of the New Zealand Land Use Suitability Database(Hamon, Mazzilli, et al., 2025). Additionally, ongoing research investigating the influence of land suitability analysis criteria on land use suitability also utilises LSAPy(Hamon, Le Roux, et al., 2025).

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